Smart Vehicle Unit - Electronics and Actuation

Independent supply | Bare UWB tag | Local command validation | Physical override

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SMART VEHICLE UNIT

E Electronic controllers | F Firmware & logic | A Actuation / regulated power | M Mechanical support

Remote MAUWBS3CA1
Pole UWB anchor
Ranging only; no control commands

UWB TWR

Makerfabs bare MaUWB tag
Regulated 3.3 V tag supply
Tag UART | UWB ranging and tag identity
Local common ground; schematic defines wiring

3.3 V DC

3.3 V Regulator for UWB Tag
Regulated logic supply for bare tag
Local common ground
Implementation and power path per schematic

Optional GPS / GNSS
GPS UART, context only, never a zone authority
Diagnostic location / event metadata
Supply and interface per schematic

M Mechanical Structure & Mounting
Compact electronics enclosure with mounting provision
Secure controller, UWB tag and demonstration circuits
Cable routing / strain relief and accessible override

Tag UART - 115200 baud / 8N1 / 3.3 V logic (matching firmware)
Tag TX to controller RX | Controller TX to tag RX
Local common ground

GPS UART, context only, never a zone authority

ESP32-S3 - Vehicle Controller
F Validation and state machine; ESP32 Wi-Fi is separate from the bare MaUWB tag radio

Pi-Hosted Local Network
2.4 GHz Wi-Fi AP
Mosquitto MQTT broker

Restrictive commands originate from the pole safety gate.

Return applied-state ACK and fault status.

No electrical cable to vehicle

Authenticated vehicle command

ACK / faults

Wi-Fi / MQTT Client
Connect to Pi 2.4 GHz Wi-Fi AP
Subscribe to intended vehicle topic
All messages through Pi Mosquitto broker

Command

Command Validation
Vehicle / tag binding; valid session; gate authorization
Application authentication; sequence / replay rejection
Command age / expiry; physical override

ACK / faults

Validated / authorized state

Acknowledgements / Diagnostics
Applied controller / output state + faults
Local watchdog and safe recovery
No separate physical load sensor specified
Relay / lamp operation needs bench validation

State

Vehicle State Machine / Output Control
Validated NHZ: coil command + fresh deadman pulses
Validated LBZ: latched HIGH/LOW lamp mode + strobe
Expiry / fault / link loss: suppression lifts locally
Stale / unknown lighting: FREEZE; no automatic HIGH

Coil + deadman

Mode + strobe

A Auxiliary Horn Control & Deadman
Horn deadman timer + MOSFET driver
Suppression needs fresh pulses AND coil command
Suppression lifts on expiry, fault, or link loss (fail-open)
Normally-closed relay: horn available when coil power is lost

NC contact state

Auxiliary Demo Horn Path
Horn button -> normally-closed relay
-> auxiliary demo sounder -> local ground
Horn available when coil power is lost
Physical override bypasses the relay
Bypass works independently of firmware

A Lamp Mode Latch & Driver
Lamp mode latch: holds HIGH/LOW mode
Stale / unknown data: FREEZE (no new strobe)
No automatic HIGH on uncertainty
MOSFET driver -> auxiliary demo lamp

Status & Diagnostics
Wi-Fi / link, zone and fault indication
Horn-button sensing and override status
Applied-state / fault reports through MQTT
Local diagnostics per schematic

Status outputs / horn-button sense / diagnostics

Override sense
Priority

A Physical Override
Physical override cuts the actuation rail
Bypasses the relay; wins even if firmware hangs
Manual HIGH/LOW lamp selection
Override sense to controller; local grounds only

Demo power

Actuation Power / Return
Automatic horn-coil / lamp actuation path
Physical override cuts automatic actuation
Manual HIGH/LOW and horn bypass remain available

SAFETY BOUNDARY
Auxiliary demo only; closed-course only. No wiring into real vehicle horn / headlamp circuits.
Physical override wins if firmware hangs; horn suppression expires locally. Uncertain lighting: no automatic HIGH.

Independent Demo Power Supply
Local supply / grounds; auxiliary demo only

Demo power

Fuse
Rating: load / wiring dependent

Master Power Switch
Feeds switched demo supply

Demo Supply & Regulation Distribution
Regulated controller / UWB tag / optional GPS supply | auxiliary demo horn path | physical override
Local grounds only; no pole-to-vehicle cable. Override cuts automatic actuation; bypass + manual HIGH/LOW remain available.

LOCAL SAFE RECOVERY | Horn deadman expiry / fault / link loss -> suppression lifts; NC relay makes horn available without coil power.
Stale / unknown lighting -> FREEZE; no automatic HIGH. Physical override cuts actuation, bypasses relay and provides manual HIGH/LOW.

LINE CONVENTIONS | Arrow direction indicates sender to receiver; matching named power nets are continuous locally.

Wired / logical data | Wireless RF | Power / voltage | Configuration | Inhibit / output control

M Mechanical structure | A Actuation & power | E Electronic controllers | F Firmware & logic | Dashed box: optional feature